



Assignment 2 - AI Improvement & Prototyping: PRD

Title	YouTube Learning State Layer — PRD
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Version	1.1.0

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1. Problem Alignment -

I.) Why? (Objective)

- 1. **For users -**

To help Indian competitive exam students (JEE, NEET, UPSC, CUET, Boards) maintain learning continuity on YouTube — so they always know what they've covered, where they left off, and what to watch next.
- 2. **For business -**

To increase YouTube's retention and session depth among one of the platform's highest-intent user segments (Indian exam-prep students), by converting one-time video viewers into returning, progressing learners, directly improving completion rates, repeat sessions, and long-term platform stickiness.

II.) How do we Measure Success ?

- 1. **Goal -**

Improve **playlist completion rate** by **20%** within **6 months** for *Indian competitive exam students* on **YouTube**.
- 2. **Success Metrics -**
 - i. North Star Metric - **Playlist Completion Rate** among active users — measured as the percentage of exam-prep playlists started that are completed meaningfully (defined as 80%+ of videos watched).
 - ii. Secondary Metrics -

- **Return rate** — % of dropped playlists that the student returns to within 7 days. Tells us if continuity is being restored after a gap.
- **Time to re-entry** — average number of days between drop-off and return. Tells us if the gap between abandonment and return is shrinking over time.
- **Restart rate** — % of returning students who restart from video 1 instead of continuing forward. Tells us if students are re-orienting successfully or still feeling lost.
- **Session depth** — average number of videos watched per study session. Tells us if students are engaging meaningfully per visit or just opening one video and leaving.

3. **Guardrail metric -**

- i. **Passive completion rate** — % of students completing playlists without meaningful watch time per video. Guards against students rushing through videos just to hit completion without learning anything.

III.) Who are the Users ?

1. **Persona -**

Aarav Sharma is a 16-year-old NEET aspirant from Lucknow who structures his days around preparation. His mornings begin with Physics Wallah lectures across Physics, Chemistry, and Biology, followed by long hours solving problems from standard reference books. He trusts structured coaching content and believes disciplined study — along with handwritten notes — is the key to cracking the exam. Yet beneath this routine is a constant pressure: NEET is a year away, the syllabus is vast, and every hour matters. When Aarav returns to YouTube to revisit a concept he half-understood weeks earlier, he often finds himself staring at a playlist he barely remembers starting — unsure what he already covered or where to resume. Faced with this confusion, he either restarts from the beginning or abandons the topic entirely. In a preparation journey where clarity and continuity are critical, YouTube frequently fails him at the exact moment he needs guidance most.

2. **Job To Be Done -**

Primary:

- When I sit down to study on YouTube, I want to feel oriented and in control of my learning path, so I can maximize every study hour in a high-stakes, time-sensitive preparation.

Secondary:

- When I return to a YouTube playlist I abandoned days ago, I want to instantly understand what I've already covered and where to re-enter, so I can continue learning without wasting time rewatching or restarting from scratch.
- When I finish watching a video, I want to know exactly which concept I just covered and what comes next in my syllabus, so I can feel like I'm progressing meaningfully rather than watching randomly.

3. **User Research -**

Validated through informal interviews with 4 Indian competitive exam students aged 16-20. Key findings: ~1 in 3 playlists abandoned mid-way, real drop-off threshold estimated at 45-50 minutes, students do not make notes despite claiming otherwise, and 'change of interest' identified as a face-saving response masking loss of context.

2. Solution Alignment

I.) Solution:

1. **Brief of the solution -**

YouTube currently has no concept of a learner's progress state — it tracks what students watch but not what they learn. Our solution is a browser extension that builds a learning state layer on top of YouTube, specifically designed for Indian competitive exam students preparing for JEE, NEET, UPSC, CUET, and Boards.

The extension maps YouTube videos to official exam syllabus, tracks concept-level progress beyond watch percentage, and generates personalized re-entry summaries when a student returns after a gap — so they always know what they've covered, where they left off, and what to watch next. Unlike YouTube's engagement-driven algorithm, the extension recommends next videos based on syllabus progression and learning gaps, not click-through potential.

The core shift this product enables is simple: from tracking what you watched to understanding what you learned.

2. **Shot-down alternatives -**

- **Mobile app** — but YouTube learning in India for such exams happens heavily on desktop/laptop during study hours.
- **YouTube feature itself** — ideal end state but requires YouTube to build it, which can be difficult to inculcate in existing set and match with the organization goal.
- **Web app companion** — possible but creates friction of switching tabs mid-study.

II.) Product flow:

A collapsible sidebar extension on YouTube. Dark mode. Minimal. Never blocks the video.

Seven screens in sequence:

S1 — Onboarding: Exam selection on first install. Single choice, one confirm.

S2 — Idle State: Sidebar shows syllabus progress ring and recent activity. Map icon accessible at all times.

- S3 — Active Watching:** Sidebar collapses to a single tracking line — topic name and unit only. Silent and minimal.
- S4 — Re-entry Overlay (Hero):** Returns after 3+ day gap — floating card below video title showing last visited, topics covered, drop-off point. Two actions: Resume or Quick Recap.
- S5 — Quick Recap:** Sidebar expands with AI summary — what was covered, where left off, what's next. Single Start Watching button.
- S6 — Syllabus Map:** Sidebar expands to full color-coded syllabus tree. Topics mapped to YouTube videos.
- S7 — Next Video Recommendation:** Overlay at video end — one recommended video based on syllabus progression. Watch Now or Skip.

III.) Important Links:

- Link to Working Prototype - [click here](#)
- Link to Product Story (Slide Deck) - [click here](#)
- Link to Prompt Details Document - [click here](#)

IV.) Pitfalls - Mitigations:

S.No.	Assumption at Risk	Failure Mode	Impact	Likelihood	Mitigation	Detection Signal	Containment/Rollback
1.)	AI can accurately classify YouTube videos to the correct exam syllabus topic.	Misclassification of videos leads to incorrect progress tracking - a student might think they've covered a topic when they haven't.	High	Medium	Launch with curated, pre-verified playlists from trusted channels before expanding to all content.	Drop in return rate and session depth; user complaints about wrong topic mapping.	Revert to manual syllabus tagging for flagged playlists; disable auto-classification until model accuracy improves
2.)	Students will actively engage with re-entry summaries when returning after a gap.	Students dismiss or ignore the summary overlay and skip straight to the video, rendering the core GenAI feature useless.	High	Medium	Keep summaries under 5 lines, non-intrusive, and dismissible in one click.	Low re-entry summary engagement rate in secondary metrics.	Make summary opt-in rather than default; test alternative re-entry triggers like a single progress bar nudge.
3.)	Students will install and continue using the extension beyond the first week.	High uninstall rate within around a week due to low perceived immediate value — students don't see benefit until they actually drop off and return.	High	High	Onboarding flow that immediately shows the student their syllabus map and progress state on first install — deliver value before first drop-off occurs.	Extension retention rate below 40% at day 7.	Trigger an early re-engagement notification within 48 hours of install showing a sample learning state to demonstrate value.

3. Dates and FAQs -

I.) Tentative Timelines:

Task	Date
Leadership approval	11-03-26
Design ready	21-03-26
Prototype testing	28-03-26
Development starts	30-03-26
Beta Launch	20-04-26

II.) FAQs:

1. What do you mean 'exactly' by learning state layer ?

Right now YouTube only knows **watch state** — you watched 47% of a video. That's it. It has no memory beyond that. A **learning state** is different. It's the answer to a deeper set of questions:

- *What concepts has this student actually covered?*
- *Where in the syllabus are they?*

- *How long ago did they last engage?*
- *What do they need to revisit versus what can they move forward from?*

2. Can you explain this layer with an example ?

Suppose you are studying Physics — NLM and Friction on YouTube from a particular playlist. A learning state layer would know you covered Newton's Laws, struggled with the friction problems based on your rewind behavior, and haven't touched the topic in 3 weeks — so it knows exactly how to bring you back in.

3. What is the major problem statement being targeted ?

Indian competitive exam students (JEE, NEET, UPSC, CUET, Boards) rely heavily on YouTube for self-directed learning, but the platform has no concept of a learner's progress state. It tracks what students *watch* but not what they *learn* — meaning when a student drops off mid-playlist and returns after a gap, they have no reliable way to understand what they've covered, where they left off, or what to study next. This absence of learning continuity causes students to either restart from scratch or abandon the topic entirely, wasting both time and motivation in a high-stakes, time-sensitive preparation context.

4. What are the numbers in the **goal** based on ?

The numbers — 20% and 6 months — are currently assumptions for the sake of this project since I don't have benchmark data.

5. Why have we considered *80%+* in the NSM ?

Because students will always skip an occasional video. 80% is a honest, realistic threshold that signals genuine progression without being impossible to achieve.

6. Why Browser Extension as a solution ?

- A browser extension sits directly on top of YouTube without requiring YouTube's cooperation or API access. It's the most practical way to add a layer to an existing product you don't own.
- The student stays on YouTube. The extension works invisibly in the background and surfaces information exactly when needed — not in a separate app they have to remember to open.
- Tools like Glasp, VideoSummarizer, and other YouTube companion tools already exist as extensions, validating the form factor for this use case.

7. How is this different from simply bookmarking a video ?

A bookmark tells you where you stopped watching. It has no memory of what you understood, no awareness of where you are in a syllabus, and no ability to tell you what comes next. The learning state layer knows which concepts you've covered, infers how well you engaged based on your behavior, and generates a contextual summary tailored to your specific gap and syllabus position. A bookmark is a pointer. This is a memory.